ARM Exceptions/Interrupts

An Exception/Interrupt is any Signal/Event/condition that needs to break/suspend/halt the normal sequential execution of Instructions.

There are 7 different exceptions in ARM7 processor. When any of these exceptions occur the operating mode of the processor changes to a corresponding privileged mode.

Exceptions are handled in the following manner.

When an exception occurs

1) The current program is **suspended**.
2) **SPSR ← CPSR**. This stores the state of flags of the original program.
3) **CPSR mode bits** are changed as the "operating mode" of the processor changes.
4) In CPSR, **T ← 0**. If the processor is in Thumb state, it changes to ARM state.
5) **LR (R14) ← PC (R15)**. Return address is stored into LR. **(Note: This does not happen in reset).**
6) Change **banked registers** as per the new operating mode.
7) **PC (R15) ← ISR address**. Load ISR address into PC and begin the ISR.

To return from the ISR

1) **CPSR ← SPSR**. This restores the original state of flags including "**T**" and **Mode bits**.
2) **PC (R15) ← LR (R14)**. Return address is restored from LR to PC.
3) **Banked Register** functions are restored per the original program mode.

7 ARM Exceptions

- ARM core is reset
- An instruction fetch fails
- Memory access fails
- Undefined instruction is encountered
- Software interrupt instruction is executed
- Fast Interrupt request is raised
- Interrupt Request is raised

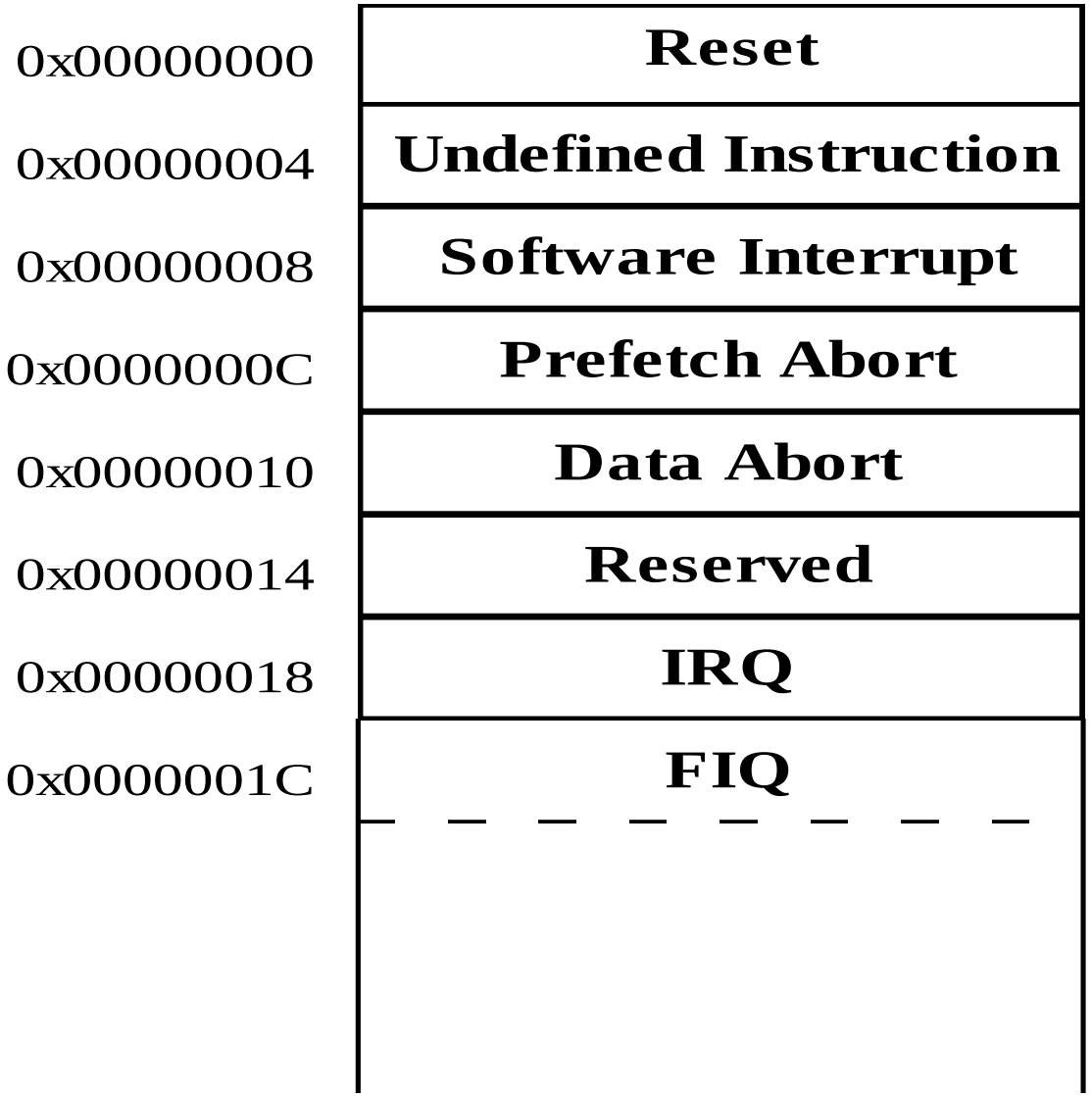
Exception Occurred	Mode Entered	ISR Address (Normal)	ISR Address (High Vectors)
Reset	Supervisor	0000 0000 H	FFFF 0000 H
Undefined Instruction	Undefined	0000 0004 H	FFFF 0004 H
Software Interrupt	Supervisor	0000 0008 H	FFFF 0008 H
Prefetch Abort	Abort	0000 000C H	FFFF 000C H
Data Abort	Abort	0000 0010 H	FFFF 0010 H
Reserved (Was earlier used in 26-bit ARM, reserved for future processors)		0000 0014 H	FFFF 0014 H
IRQ (Normal Interrupt)	IRQ	0000 0018 H	FFFF 0018 H
FIQ (Fast Interrupt)	FIQ	0000 001C H	FFFF 001C H

Multiple Exceptions can occur simultaneously.

In such case the following priority order is followed to decide which interrupt would be serviced first.

EXCEPTION	PRIORITY	
Reset	1	*(Highest)*
Data Abort	2	
FIQ	3	
IRQ	4	
Prefetch Abort	5	
Undefined Instruction	6	*(Lowest)*
Software Interrupt		

Vector Table



Exception Priorities

Priority		Exception
Highest	1	Reset
	2	Data Abort
	3	FIQ
	4	IRQ
	5	Prefetch Abort
Lowest	6	Undefined Instruction SWI

Reset

- This exception is caused by an active low signal on the **nReset** pin.
- PC becomes 0000 0000H.
- From this location the BIOS program (Booting / Monitor program) is executed in supervisor mode.
- It is the highest priority exception (Level 1).
- It disables all interrupts by making **I ← 1** and **F ← 1** in CPSR.

Undefined Instruction

- This exception occurs when an undefined instruction is encountered.
- This typically happens when a co-processor instruction is written in the program but the co-processor is absent.
- It disables IRQ by making I ← 1 in CPSR.

Software Interrupt

- This interrupt can be invoked by the programmer, by writing SWI instruction in the program.
- The operating mode changes to Supervisor mode.
- This can be used by user programs, to invoke OS tasks that cannot be performed normally in User mode.
- Its priority level is same as Undefined Exception (Level 6).
- This is because both these interrupts cannot occur at the same time.
- It disables IRQ by making I ← 1 in CPSR.

Prefetch Abort

- This exception occurs when an attempt is made to fetch an instruction from an invalid memory location.
- This may happen due to two cases. Either the locations is not physically mapped (all systems may not actually connect the entire 4GB memory) or if the program does not have permission to access the desired location.
- It disables IRQ by making I ← 1 in CPSR.

Data Abort

- This exception occurs when an attempt is made to perform a data operation (read or write) at an invalid memory location.
- This may happen due to the same reasons as Prefetch abort but also additionally if a "data write" attempt is made on a "read-only" location.
- It disables IRQ by making I ← 1 in CPSR.

IRQ

- This interrupt occurs when an active low signal is applied to nIRQ pin.
- It is treated as a lower priority hardware interrupt.
- It disables further IRQ interrupts by making I ← 1 in CPSR.

FIQ

- This interrupt occurs when an active low signal is applied to nFIQ pin.
- It is treated as a higher priority hardware interrupt.
- It disables further IRQ and FIQ interrupts by making I ← 1 & F ← 1 in CPSR.

High Vectors

- In some versions of ARM, Interrupt Vectors can be placed in the beginning of the memory map as seen above or towards the end of the memory map (FFFF XXXXH) called as High Vectors.
- If a "System Control" coprocessor is used, then in its control register, there is a bit "V", that decides whether normal vectors (V=0) or High Vectors (V=1) should be used. By Default Normal vectors are used so ISR addresses begin from 0000 0000H onwards.